

Referential Regimes: Transformation-Invariant Identity for Neutral Substrates

Denise M. Case

Northwest Missouri State University, Computer Science and
Information Systems, Maryville, MO, USA

Abstract

Data systems increasingly operate under persistent legal, political, and analytic disagreement, where no single interpretive authority can be assumed. A *neutral substrate* provides stable shared reference without requiring agreement about causal or normative interpretation: it fixes reference structurally and leaves interpretation to extension layers, following the neutrality-by-design constraint developed in *Neutral Substrates* (Case 2026).

This paper derives the identity structure such a substrate requires, using a small transformation algebra rather than a list of kinds. We treat identity as transformation-invariance: a referent is individuated by the admissible operations that preserve it. Once the routine operations that record systems perform are fixed, including re-expression, annotation, refinement, decomposition, reassignment, forking, and provenance extension, individuation becomes a structural calculation.

That calculation exposes a mismatch between reference kinds and identity regimes. An accountability substrate must refer to six kinds of thing: obligation-bearers, rules, occurrences, scopes, records, and plain referents. Three of those kinds are assigned more than one admissible basis. A scope may be fixed by its extension or by its structure; a rule

by its content or by its structure; a plain referent by its locus or by its object. A single routine operation separates each pair: decomposition separates the two scope readings, refinement the two rule readings, and forking the two plain-referent readings.

Counting reference kinds gives six; transformation analysis forces at least nine identity regimes. The three additional regimes add no new kind to the inventory. They follow from how identity behaves under operations the substrate cannot avoid performing, and any attempt to collapse them reappears as a hidden regime that persistent disagreement cannot hold fixed.

The result is a conditional lower bound, monotone in the relevant sense: a richer transformation basis may leave the bound unchanged or force further splits, but it cannot erase a distinction already witnessed here. Its content is diagnostic. The additional regimes correspond to distinctions often handled by modeling convention, but such conventions cannot substitute for regime-level identity commitments under persistent disagreement.

Keywords: formal ontology; identity and persistence; neutral substrates; extension stability; accountability; causal commitment; normative commitment

1 Introduction

Modern accountability systems increasingly operate under legal, institutional, and analytic disagreement. A shared data substrate cannot assume a single interpretation, and it must preserve stable reference as records are amended, forked, refined, decomposed, aggregated, and reclassified. This paper studies the identity conditions required by that setting.

The analysis treats identity as transformation-invariance. An identity regime states the operations that preserve sameness for a carrier under a declared identity basis. Once the ordinary operations of record systems are fixed, individuation becomes a structural question: the identity basis preserved across

those operations determines sameness. A system may collapse a distinction at the substrate level and recover it through a role, flag, contextual predicate, workflow state, schema convention, or application-level discriminator. That recovered distinction is a hidden regime.

For accountability substrates the inventory has six carrier kinds: obligation-bearers, rules, occurrences, scopes, records, and plain referents. Identity fixes sameness under transformation, relative to a declared basis. Three carrier kinds admit more than one basis in the settings analyzed here. A scope may be individuated by covered cases or by internal organization. A rule may be individuated by imposed requirements or by textual structure. A plain referent may be individuated by its location or by the object occupying it.

These alternatives are separated by ordinary operations. Decomposition separates extension-fixed from structure-fixed scopes. Refinement separates content-fixed from structure-fixed rules. Forking separates locus-fixed from object-fixed plain referents. The same pair of representations may be equivalent under one basis and distinct under another. When admissible frameworks require both bases, a neutral substrate must preserve both.

The lower-bound core follows from that structure. The six carrier kinds provide a floor of six regimes. The three witnessed splits add one regime each. Thus a neutral accountability substrate satisfying the stated assumptions, and supporting those basis-plural cases, requires at least nine identity regimes. The additional regimes arise within kinds already named, so an analysis that stops at naming kinds will undercount. Collapsing a split either changes the transformations that preserve identity or reintroduces the distinction through a hidden regime.

The result is conditional. It holds under the assumptions fixed in Section 3 for substrates in which the witnessed basis pluralities are in play. Adding transformations may force further splits; it cannot erase a distinction witnessed by the current basis. This paper addresses identity. Conformance checking is left to the companion *Accountable Records* paper.

The remainder of the paper proceeds as follows. Section 2 imports definitions the proofs use and states one new primitive. Section 3 fixes the operating conditions. Section 4 states the reference inventory. Section 5 develops the transformation algebra. Section 6 collects the nine core regimes. Section 7 proves the lower bound. Section 8 establishes core regime assignment. Section 9 exhibits the regimes in the diagnostic settings. Section 10 situates the result in prior work. Section 11 states limits. Section 12 concludes. Appendix A provides the pairwise separation table.

2 Background: Neutral Substrates

This paper builds on *Neutral Substrates* (Case 2026). That paper defines the design constraint for a shared representational base that must remain usable under persistent interpretive disagreement. This paper analyzes one aspect of that constraint: the identity and persistence structure needed for stable reference to survive ordinary transformations of records.

The definitions in this section are inherited from that work. They are reproduced here to make the dependency clear. Stable SE-100 labels are preserved for imported definitions.

2.1 Inherited Definitions

Definition 2.1 (Substrate). A *substrate* $\mathcal{S}$ is a shared representational base providing stable reference for entities, occurrences, and institutional artifacts across a class of admissible interpretive frameworks $\mathbb{F}$. Stable reference consists of individuation, co-reference, and persistence.

A substrate is not the whole record system. It is the foundational layer for later interpretive claims, including causal or normative interpretations, evidentiary assessments, explanatory accounts, and institutional determinations. Its purpose is to make shared reference possible without forcing agreement

about every interpretation of the referent. A substrate may identify an event, entity, decision, document, instrument, rule, policy, observation, or institutional artifact, and it may preserve identifiers, timestamps, provenance, and persistence relations among them. Identifying the relevant event, decision, rule, or institutional artifact does not commit the substrate to claims about causation, justification, or applicability.

Definition 2.2 (Referential Regime). A *referential regime* is the triple of

1. individuation conditions,
2. co-reference conditions, and
3. persistence conditions

used by a substrate to fix and track entities, occurrences, and institutional artifacts.

Individuation conditions determine when something counts as one thing rather than another. Co-reference conditions determine when two references refer to the same thing. Persistence conditions determine when something remains the same thing across time, transformation, revision, or institutional change. The referential regime allows a substrate to support shared use: if parties cannot share enough reference to identify the referent under interpretation, they cannot reliably disagree about causes, norms, explanations, or obligations concerning that referent.

Definition 2.3 (Referential Commitments). Let $\mathcal{S}_{\text{ref}}$ denote the *referential commitments* of $\mathcal{S}$. These are the substrate-layer commitments fixed by its referential regime, including identifiers; the typing of entities, occurrences, and institutional artifacts; timestamps; provenance; and referential relations among them.

Neutral Substrates also defines permitted attribution propositions, object-level causal or normative commitments, interpretive non-commitment, extension

stability, and neutrality by design. Those definitions support its central *constraint*: under the stated contestability and common-ground assumptions, a substrate’s neutrality is guaranteed at design time exactly when its foundational layer is restricted to referential commitments and permitted attribution propositions. Specifically, the substrate must not make object-level causal or normative commitments. Causal and normative content may appear through attributed assertions, but the substrate commits to the attribution, not to the asserted causal or normative proposition.

2.2 Contribution

Neutral Substrates establishes the design constraint: a neutral substrate requires stable reference while leaving causal and normative interpretation to admissible extensions. This paper analyzes the identity structure needed to preserve stable reference as substrate records are transformed in ordinary use. Records may be re-expressed, annotated, refined, decomposed, aggregated, reassigned, substituted, forked, extended with provenance, or evolved through state change. Those operations place pressure on individuation, co-reference, and persistence. This paper specializes the referential-regime component of that constraint by deriving the lower-bound identity structure it imposes.

Definition 2.4 (Transformation-Invariant Identity). Within a neutral substrate, an *identity regime* specifies how a referent’s identity basis behaves under admissible transformations. For each transformation, the regime specifies whether the transformation preserves identity, breaks identity, is identity-neutral, or is inapplicable. A referent is *transformation-invariant* under transformations that do not break its declared identity basis.

Section 5 makes Definition 2.4 operational as a classification over a fixed transformation basis. The rest of the paper computes the resulting lower bound.

3 Formal Setting

The results in this paper are conditional on the assumptions fixed here. These assumptions specify the operating conditions under which the neutral-substrate constraint of Section 2 is applied. The goal is to identify the lower-bound identity structure required of a neutral accountability substrate under those conditions.

3.1 Persistent Disagreement

We assume environments in which disagreement is persistent and extensions are uncoordinated. Extensions may introduce incompatible causal accounts, incompatible normative attributions, and incompatible interpretive classifications. They cannot be forced to converge. Substrate revision may not be used to resolve interpretive conflict. Consequently, substrate-level admissibility must not depend on agreement among extensions. Neutrality must be maintained without recourse to convergence.

Persistent disagreement is not mere temporary uncertainty. It is the condition in which multiple admissible frameworks may remain simultaneously usable while disagreeing about causation, responsibility, justification, applicability, or normative significance. A neutral substrate must therefore support shared reference without embedding any one framework’s interpretation as the substrate-level account.

3.2 Stable Reference in Strong Form

The persistence component of the referential regime is inherited from *Neutral Substrates*. Here we fix the demanding case in which persistence must be evaluated under ordinary transformations of records. Identity conditions must not depend on interpretive convergence. An entity that must change its ontological status to enter a new relationship has not been stably referred to.

A stability-critical distinction must be represented in the substrate, rather than carried by a role, flag, or contextual predicate. Such devices rely on downstream conventions and can therefore permit divergent judgments under persistent disagreement.

This is a deliberately strong setting. Engineering systems often manage identity pressure through negotiated, local, or context-relative schemes, which may be useful in practice but are not the case analyzed here. This paper studies the case in which stable reference must be available as common ground before interpretive agreement is available.

3.3 Minimality

We minimize referential structure subject to the above constraints. Minimality is measured by the number of identity regimes needed to preserve neutrality and stable reference. A reduction is admissible when it preserves the relevant identity distinctions without relocating them into roles, flags, or other discriminators.

A reduction counts as genuine when it does not reintroduce a collapsed distinction through a role, flag, contextual predicate, workflow state, schema convention, or application-level discriminator. The criterion is analytic. It identifies the regime structure required under the stated constraints. It does not recommend use of the core set beyond the neutral-substrate conditions analyzed here.

3.4 Transformation Basis

Identity pressure appears when records are transformed. We therefore fix a finite transformation basis

$$\mathcal{F} = \{\text{RE}, \text{AN}, \text{RF}, \text{AD}, \text{RC}, \text{RA}, \text{SU}, \text{BF}, \text{PV}, \text{SE}\}.$$

The basis is not claimed to be complete for all possible record systems. It is a representative basis of ordinary operations in provenance, versioning, institutional recordkeeping, and data-governance practice. The lower-bound result is conditional on this basis. Adding further transformations cannot collapse distinctions already witnessed by this basis. It may leave the lower bound unchanged or force further splits.

Definition 3.1 (Transformation basis). The transformation basis $\mathcal{F}$ consists of the following admissible operation families:

- (a) RE, re-expression: change of representation format while preserving the represented content;
- (b) AN, annotation: addition of non-identity-bearing information;
- (c) RF, refinement: increase in descriptive, structural, or normative detail;
- (d) AD, aggregation or decomposition: whole-preserving restructuring into coarser or finer components;
- (e) RC, re-contextualization: change in applicability context;
- (f) RA, reassignment: change in association, bearer, or assignment relation;
- (g) SU, substitution: replacement of the identity carrier;
- (h) BF, branch or fork: divergence into multiple continuations;
- (i) PV, provenance extension: addition of historical, derivational, or evidentiary metadata;
- (j) SE, state evolution: change in state of an enduring referent over time.

The transformation families are typed operation families. A transformation is evaluated relative to the carrier-basis pair to which it is applied. In particular, AD names aggregation or decomposition that preserves the represented whole while changing its component structure. For scopes, this means preserving the covered cases while changing the internal scope organization. Operations that add or remove covered cases fall outside AD. The transformation basis is representative rather than complete, as stated in Definition 3.1. By Corollary 7.3, adding further transformations cannot lower the bound. A transformation may break identity when it restructures identity-relevant components. It may remain non-breaking when it restructures a non-identity-

bearing description.

3.5 Classification of Transformations

A regime classifies each applicable transformation by how it affects the declared identity basis. We use four classifications.

Definition 3.2 (Transformation classification). For a regime τ and transformation family $f \in \mathcal{F}$, the classification

$$\text{Class}_\tau(f) \in \{\text{PRS}, \text{BRK}, \text{NEU}, \text{N/A}\}$$

has the following meaning:

- (a) PRS: the transformation is identity-preserving for τ ;
- (b) BRK: the transformation is identity-breaking for τ ;
- (c) NEU: the transformation is identity-neutral for τ ;
- (d) N/A: the transformation is inapplicable to τ .

The distinction between PRS and NEU matters. A PRS transformation preserves identity in a way that is relevant to the regime’s identity basis. An NEU transformation does not bear on that identity basis. Both are non-breaking. A transformation breaks identity when its classification is BRK. Thus NEU does not mean that the operation is ignored by the system. It means that the operation does not alter identity under the regime being considered. A fourth value, N/A, marks a transformation that cannot arise for the regime’s carrier at all, as distinct from one that arises but leaves identity untouched.

Definition 3.3 (Non-breaking transformation). A transformation family f is *non-breaking* for regime τ when

$$\text{Class}_\tau(f) \in \{\text{PRS}, \text{NEU}\}.$$

It is *breaking* for τ when

$$\text{Class}_\tau(f) = \text{BRK}.$$

3.6 Identity Regimes

The new primitive of this paper is the identity regime. An identity regime is not an ontological kind. It is a transformation profile over an identity basis.

Definition 3.4 (Identity regime). An *identity regime* τ consists of an identity basis together with a classification function

$$\text{Class}_\tau : \mathcal{F} \rightarrow \{\text{PRS}, \text{BRK}, \text{NEU}, \text{N/A}\}.$$

The identity basis specifies sameness for the referent. The classification function states the effect of each transformation family under that basis.

Definition 3.5 (Induced identity relation). Let τ be an identity regime. The induced identity relation $\sim_\tau$ is the equivalence relation generated by transformations that are non-breaking for τ . Thus $r \sim_\tau s$ means that r and s are representations of the same referent under regime τ .

This definition makes identity transformation-relative. The same pair of representations may coincide under one regime and diverge under another. The substrate must declare which identity basis is being preserved.

3.7 Basis Plurality

A carrier kind exhibits basis plurality when admissible frameworks require stable reference to that carrier under more than one identity basis. Basis plurality is stronger than the mere availability of multiple descriptions. It means that the substrate must support frameworks that preserve different sameness conditions for the same carrier kind.

Definition 3.6 (Basis plurality). A carrier kind has *basis plurality* when admissible frameworks require stable reference to that carrier kind under more than one identity basis. A neutral substrate serving those frameworks must preserve each required basis without relocating the distinction into a hidden regime.

The lower-bound proof uses three witnessed cases of basis plurality. Plain referents may be tracked by locus or object, scopes by extension or structure, and rules by content or structure. Other admissible bases may exist. When both witnessed bases are in play, one regime cannot serve both without changing identity behavior or recovering the distinction elsewhere.

3.8 Hidden Regimes

A hidden regime is a collapsed identity distinction recovered outside the declared substrate structure. Hidden regimes obstruct genuine minimality. They allow a system to appear to use fewer identity regimes while relying on downstream discriminators to perform the missing identity work.

Definition 3.7 (Hidden regime). A *hidden regime* is a stability-critical distinction that is not represented as an identity regime in the substrate, but is instead recovered through a role, flag, contextual predicate, workflow state, schema convention, or application-level discriminator.

Hidden regimes violate the neutral-substrate discipline because their application is not fixed by the substrate. They require agreement about how the discriminator should be interpreted. Under persistent disagreement, that agreement cannot be assumed. A collapse repaired through a hidden regime is therefore not a genuine reduction in declared referential structure.

3.9 Scope of the Lower Bound

The lower bound proved below is conditional. It holds for neutral accountability substrates satisfying the assumptions of this section, using the transformation basis of Definition 3.1, and supporting the witnessed basis pluralities of Definition 3.6.

Under those conditions, fewer than nine regimes cannot preserve stable reference without changing transformation behavior or reintroducing a hidden regime.

The result is open-ended upward but fixed downward. The carrier kinds, identity bases, and transformation families used here are sufficient to witness the core. Further regimes may be required in richer deployments. That possibility does not affect the lower bound proved here.

The next section states the reference inventory. The inventory supplies the referents that must remain addressable. The transformation basis supplied here determines which of those referents can share an identity regime.

4 Reference Inventory

Before analyzing identity, we fix the carrier inventory. An accountability substrate must let one state, inspect, and compare claims about who is responsible, which authority governs, what happened, where it applies, what was recorded, and what was acted upon. It must do this without embedding causal or normative content at the substrate level.

The inventory is modest. It is not a theorem, an upper ontology, or a complete domain ontology. It states the carrier kinds required by the accountability setting analyzed here. Those carriers are the referents that must remain addressable under the assumptions of Section 3.

The inventory supplies identity carriers but does not determine identity

regimes. A reference kind identifies the referents that must remain addressable. An identity regime specifies sameness for a carrier under transformation. The lower-bound proof depends on that distinction.

4.1 Reference Kinds and Identity Carriers

Definition 4.1 (Reference kind). A *reference kind* is a class of referents that a neutral accountability substrate must be able to identify, track, and distinguish from other referents without making object-level causal or normative commitments.

Definition 4.2 (Identity carrier). An *identity carrier* is a referential position in the substrate to which an identity regime may be assigned. It is the object whose sameness, difference, and persistence are evaluated under transformation.

Reference kinds supply identity carriers. They do not themselves determine identity regimes. A single reference kind may support more than one identity regime when different identity bases respond differently to the same transformation. Conversely, the fact that two carriers are both represented in the substrate does not imply that they share an identity regime. Regime assignment depends on transformation behavior, not on the name of the kind.

The inventory used in this paper is:

Reference kind	Required referent
Obligation-bearer	a party that can bear responsibility across the occurrences discharging, violating, or contesting that responsibility
Rule	a governing normative structure, distinct from the acts that enact, amend, apply, or challenge it
Occurrence	a happening individuated by temporal realization and provenance
Scope	the applicability domain for a rule, including places, parties, periods, procedures, classifications, or conditions
Record	a descriptive indicator or record that carries reference without asserting causal or normative truth
Plain referent	a thing acted upon, regulated, located, transferred, or observed that neither bears obligations nor grounds authority as such

These six reference kinds are familiar from foundational ontology, legal ontology, provenance, and institutional recordkeeping. The paper does not claim novelty for the inventory itself. The inventory matters because it supplies the carriers on which the later identity analysis operates.

4.2 Obligation-Bearers

An accountability substrate must support reference to obligation-bearing entities. An obligation-bearer is a party that can bear responsibility, duty, liability, authority, or accountability across occurrences in which those statuses are fulfilled, violated, transferred, contested, or discharged. The substrate need not decide whether a party is responsible in a particular case. It must preserve reference to the party that a framework may treat as responsible.

In multi-jurisdiction regulatory reporting, a corporate entity may be treated differently across jurisdictions. One jurisdiction may individuate responsibility at the parent-entity level. Another may individuate it at the subsidiary, facility, role, or delegated-agent level. A neutral substrate need not settle which legal theory governs. It must still allow the relevant bearer to remain referable across those competing frameworks.

Obligation-bearers cannot be reduced to the occurrences in which obligations are discharged. The same bearer may participate in many occurrences. The same occurrence may involve several bearers. Collapsing bearer into occurrence destroys the ability to compare responsibility across time and across frameworks.

4.3 Rules

An accountability substrate must support reference to governing rules or normative structures. A rule is a structure that may be enacted, amended, delegated, interpreted, applied, violated, or repealed. The substrate need not assert that the rule is valid, justified, binding, or correctly interpreted. It must preserve reference to the rule as the object of such attributed claims.

In legislative history tracking, a statute must remain referable independently of the legislative act that enacted it, the amendment that altered it, the agency action that applied it, and the judicial decision that interpreted it. A statute can be contested while still remaining the same referent for purposes of citation, comparison, and provenance. If the rule is collapsed into its enactment or application occurrence, then later disagreement about the occurrence changes reference to the rule itself.

Rules also preview one of the split cases below. A rule may be individuated by imposed requirements or by textual or internal structure. Therefore, the reference kind “rule” does not fix a single identity regime; transformation behavior does.

4.4 Occurrences

An accountability substrate must support reference to *occurrences*. An occurrence is a happening individuated by temporal realization and provenance. Illustrative examples include filings, inspections, votes, transfers, transactions, observations, decisions, enactments, amendments, and enforcement actions. The substrate must preserve reference to the occurrence as something that happened, independently of how competing frameworks evaluate it.

In multi-jurisdiction regulatory reporting, two jurisdictions may disagree about whether a filing constitutes compliance. They must nevertheless be able to refer to the same filing. In legislative history tracking, parties may disagree about the legal significance of an enactment. They must nevertheless be able to refer to the same enactment occurrence. In provenance analysis, institutions may disagree about the meaning or reliability of a measurement. They must nevertheless be able to refer to the same measurement occurrence.

Occurrences cannot be reduced to obligation-bearers, rules, records, or scopes. They are time-indexed referents. They may instantiate, affect, or be described by other carriers, but they are not identical with them.

4.5 Scopes

An accountability substrate must support reference to scopes. A scope is the applicability domain of a rule, including places, parties, periods, procedures, classifications, or conditions. Scopes may be territorial, institutional, temporal, procedural, classificatory, jurisdictional, or conditional. They may be nested, overlapping, partially coincident, or contested.

In multi-jurisdiction regulatory reporting, the same rule may apply to different entities, places, facilities, transactions, or reporting periods under different frameworks. If scope is represented as an attribute of the rule, then a change in applicability can appear to change the identity of the rule. If scope is represented as a property of obligation-bearers, then the domain of

applicability cannot be compared independently of the parties within it. A neutral substrate needs scope as a separate referent.

Scopes also preview one of the split cases below. A scope may be individuated extensionally, by covered cases, or structurally, by internal jurisdictional or applicability organization. Decomposition can preserve one basis while changing the other. Therefore, the reference kind “scope” does not fix a single identity regime; transformation behavior does.

4.6 Records

An accountability substrate must support reference to descriptive records. A record is a descriptive indicator, document, dataset entry, observation record, provenance item, or other referent that can be cited, transferred, compared, annotated, or analyzed without the substrate asserting a causal or normative claim. The substrate may record that a source asserts a claim. It does not thereby assert the claim.

In cross-institutional provenance, a record may be produced by one institution, received by another, annotated by a third, and reanalyzed by a fourth. The institutions may disagree about what the record shows or whether it supports a conclusion. They must nevertheless be able to refer to the same record through custody, annotation, and reuse.

Records cannot be reduced to the entities, occurrences, rules, or claims they describe. Stable reference to records is required independently of those referents, including when a record supports an attributed causal or normative claim.

4.7 Plain Referents

An accountability substrate must also support reference to plain referents. A plain referent is a thing acted upon, regulated, located, transferred, observed, measured, or described that neither bears obligations nor grounds authority.

Examples include facilities, parcels, instruments, devices, environmental sites, physical objects, infrastructure components, samples, and assets. The category is intentionally modest. It captures referents that are important to accountability without already being obligation-bearers, rules, occurrences, scopes, or records.

Plain referents are needed because many accountability questions concern things that are acted upon or regulated. A regulated facility may be inspected. A parcel may fall within a jurisdictional scope. An instrument may produce a measurement record. A sample may be transferred across institutions. The substrate must preserve reference to the thing without deciding which causal or normative interpretation applies to it.

Plain referents also preview one of the split cases below. A referent may be individuated by locus, such as the site or place that remains fixed. It may also be individuated by the object occupying that locus. Forking can separate those bases. Therefore, the reference kind “plain referent” does not fix a single identity regime; transformation behavior does.

4.8 Summary

The inventory supplies six reference kinds: obligation-bearer, rule, occurrence, scope, record, and plain referent. These kinds are the carriers for the lower-bound construction and do not determine identity regimes.

The inventory identifies the required referents. The transformation analysis determines their behavior under ordinary record operations. The core is derived after applying those operations to the carrier-basis pairs.

The next section applies the transformation basis of Section 3 to these six carriers.

5 Transformation-Invariant Identity

The previous section fixed the referents the substrate must preserve. This section analyzes the conditions under which two representations of such a referent count as the same entity. The analysis treats identity as *transformation-invariance*: an identity regime is determined by the behavior of a declared identity basis under the transformation basis fixed in Section 3.

The governing idea is familiar from formal ontology: categories are individuated by identity conditions, not by classificatory convenience, and properties that a thing can gain or lose cannot carry identity (Guarino 1999; Guarino and Welty 2002). This paper applies that idea at the substrate level and in operational form. A referent remains the same under a regime when the transformations applied to its representations do not break the identity basis for that regime. Two representations may be the same under one regime and different under another. A substrate that leaves the regime implicit has not fixed reference. It has deferred the identity question to a later interpretive choice. Under persistent disagreement, that choice cannot be assumed shared.

5.1 Identity Bases

Definition 5.1 (Identity basis). An *identity basis* is the feature, structure, role in reference, or persistence condition whose preservation constitutes remaining the same referent under a given identity regime.

An identity basis is the condition a regime tracks through transformation. The lower-bound construction uses responsibility-bearing identity for obligation-bearers, temporal-realization and provenance identity for occurrences, and descriptive-record identity for records. It also uses three witnessed split cases. A rule may be content-fixed or structure-fixed. A scope may be extension-fixed or structure-fixed. A plain referent may be locus-fixed or object-fixed.

The inventory supplies carriers. The identity basis determines sameness for a carrier. The transformation classification determines whether that basis is preserved or broken. The lower-bound argument begins when two bases for the same carrier kind classify an ordinary transformation differently.

5.2 Profiles and Collapse

The formal setting already defined identity regimes as classification profiles over the transformation basis. We now use the induced identity relation $\sim_\tau$ to compare regimes.

Definition 5.2 (Collapse). Two identity regimes τ and τ' *collapse* when they induce the same identity relation:

$$\sim_\tau = \sim_{\tau'} .$$

They are *distinct* when some pair of representations is identified under one regime and separated under the other.

Definition 5.3 (Underdetermined kind). A reference kind is an *underdetermined kind* when the kind admits more than one identity basis and those bases classify some transformation family differently. Equivalently, one admissible basis for the kind classifies a transformation as non-breaking, while another admissible basis for the same kind classifies that transformation as breaking.

Underdetermination is the transformation-level counterpart of basis plurality: basis plurality requires support for both bases, while underdetermination shows that one regime cannot support them both. When basis plurality is in play for an underdetermined kind, no single regime profile serves all required bases. The kind must be represented through distinct identity regimes. The split is forced by transformation behavior, not by the name of the kind.

5.3 Carrier Kinds Not Split in the Core Construction

Three reference kinds contribute one carrier-basis pair each to the core. They are not split by the witness operations used in this construction: obligation-bearers, occurrences, and records.

An *obligation-bearer* is individuated by persistence as the party that can bear responsibility, duty, liability, authority, or accountability. Re-expression, annotation, re-contextualization, reassignment, branching, provenance extension, and state evolution do not alter that identity basis. Refinement and decomposition preserve identity when they add detail or decompose descriptions of the bearer without replacing the bearer. Substitution breaks identity because it replaces the bearer.

An *occurrence* is individuated by temporal realization and provenance. Re-expression, annotation, refinement, re-contextualization, and provenance extension do not replace the occurrence. State evolution preserves identity when it tracks phases or stages of the same occurrence. Decomposition breaks identity when it produces sub-occurrences with their own temporal individuation. Substitution breaks identity because it replaces the occurrence. Branching and reassignment are inapplicable to a fixed realization event.

A *record* is individuated by descriptive persistence without substrate-level causal or normative commitment. Annotation, refinement, decomposition, branching, and provenance extension preserve record identity because the record survives enrichment, derivation, copying, and descriptive elaboration. Re-expression, re-contextualization, reassignment, and state evolution are identity-neutral for the record as such. Substitution breaks identity because it replaces the record.

These three carrier-basis pairs are included once in the core. We name their regimes:

OBL, OCC, REC.

5.4 Witnessed Split Kinds

Three reference kinds have witnessed split pairs in the lower-bound construction. Each admits at least the two admissible readings used here, and an ordinary transformation separates those readings. The pattern is the same in all three cases. A kind admits two bases B_1 and B_2 . A transformation f is non-breaking for B_1 and breaking for B_2 . When admissible frameworks require both bases, the kind cannot be governed by a single identity regime.

Lemma 5.4 (Split pattern). *Let K be a reference kind with two admissible identity bases B_1 and B_2 . Suppose there is a transformation family $f \in \mathcal{F}$ and representations r, s such that f preserves identity under B_1 but breaks identity under B_2 . If a neutral substrate must support both bases for K , then K requires at least two distinct identity regimes.*

Proof. If one regime governed both bases, it would assign a single classification to f for the kind. But the B_1 reading requires f to be non-breaking, while the B_2 reading requires f to be breaking. No single classification can satisfy both readings. Thus the kind is underdetermined at f .

Applying f to the witnessing representations identifies them under B_1 and separates them under B_2 . The two bases therefore induce distinct identity relations. When both bases must be supported, the substrate requires at least two distinct identity regimes for K . $\square$

5.5 Scope Under Decomposition

A scope may be individuated by its extension or by its structure. An extension-fixed scope is individuated by the cases, persons, places, entities, or conditions it covers. A structure-fixed scope is individuated by the internal organization through which that coverage is represented.

Proposition 5.5 (Scope Underdetermination). *The scope reference kind is underdetermined at aggregation/decomposition AD.*

Proof. Let r be a scope fixed by extension. Let s be a decomposition of r into subsopes whose union covers the same cases. The extension is preserved, so AD is non-breaking for the extension-fixed reading. Let r' be a scope fixed by structure. Let s' be a decomposition of r' into a finer internal partition. The structure is changed, so AD is breaking for the structure-fixed reading. Both readings are admissible scopes. No single classification of AD can serve both. By Lemma 5.4, when both bases must be supported, scope requires at least two identity regimes. $\square$

This yields:

SCOPE-E, SCOPE-S.

SCOPE-E is the extension-fixed scope regime. SCOPE-S is the structure-fixed scope regime. After decomposition, two representations can apply to the same cases while no longer being the same scope structure. State evolution is inapplicable to both scope regimes: a scope is an applicability domain, not an enduring referent that passes through internal states.

5.6 Rule Under Refinement

A rule may be individuated by its content or by its structure. A content-fixed rule is individuated by its requirements, permissions, prohibitions, authorizations, or definitions. A structure-fixed rule is individuated by the organization of the text, sections, clauses, delegations, or internal rule structure through which that content is expressed.

Proposition 5.6 (Rule Underdetermination). *The rule reference kind is underdetermined at refinement RF.*

Proof. Let r be a rule fixed by normative content. Let s be a refinement of r that preserves the rule's requirements while adding detail or reorganizing expression. The content is preserved, so RF is non-breaking for the content-fixed reading. Let r' be a rule fixed by textual or internal structure. Let s' be

a refinement of r' that changes that structure. The structure is changed, so RF is breaking for the structure-fixed reading. Both readings are admissible rules. No single classification of RF can serve both. By Lemma 5.4, when both bases must be supported, rule requires at least two identity regimes. $\square$

This yields:

RULE-C, RULE-S.

RULE-C is the content-fixed rule regime. RULE-S is the structure-fixed rule regime. After refinement, two representations can require the same thing while no longer being the same text or rule structure. State evolution is inapplicable to both rule regimes: a rule is a normative structure, not an enduring referent that passes through internal states. Changes to rule content are amendments or refinements, not state transitions of a persisting object.

5.7 Plain Referent Under Forking

A plain referent may be individuated by locus or by object. A locus-fixed referent is individuated by the place, site, slot, location, or institutional position that remains fixed. An object-fixed referent is individuated by the object, asset, instrument, sample, device, parcel, or thing occupying that locus.

Proposition 5.7 (Plain-referent Underdetermination). *The plain-referent kind is underdetermined at branch/fork BF.*

Proof. Let r be a plain referent fixed by locus. Let s be a forked continuation over the same locus. The locus is preserved, so BF is non-breaking for the locus-fixed reading. Let r' be a plain referent fixed by object. Let s' be a fork into non-identical object continuations. The object identity is not preserved across both continuations, so BF is breaking for the object-fixed reading. Both readings are admissible plain referents. No single classification of BF

can serve both. By Lemma 5.4, when both bases must be supported, plain referents require at least two identity regimes. $\square$

This yields:

$$\text{LOC}, \quad \text{OBJ}.$$

LOC is the locus-fixed plain-referent regime. OBJ is the object-fixed plain-referent regime. After a fork, two representations can be the same place while no longer being the same object. State evolution applies to both plain-referent regimes and preserves identity: a locus or an object is an enduring referent that persists through changes in its state.

5.8 Result of the Algebra

The reference inventory supplied six carrier kinds. Three carrier-basis pairs contribute one regime each:

$$\text{OBL}, \quad \text{OCC}, \quad \text{REC}.$$

Three carrier kinds have witnessed split pairs:

$$\text{LOC}, \text{OBJ}, \quad \text{SCOPE-E}, \text{SCOPE-S}, \quad \text{RULE-C}, \text{RULE-S}.$$

Thus the transformation analysis yields the nine-regime lower-bound core:

$$\begin{aligned} &\text{OBL}, \quad \text{OCC}, \quad \text{REC}, \quad \text{LOC}, \quad \text{OBJ}, \\ &\text{SCOPE-E}, \quad \text{SCOPE-S}, \quad \text{RULE-C}, \quad \text{RULE-S}. \end{aligned}$$

The additional regimes arise within carrier kinds already named. They are identity bases separated by ordinary transformations. This is why counting carrier kinds undercounts the structure needed for stable reference. The next section collects the nine core regimes in a single classification table.

6 The Nine-Regime Lower-Bound Core

The transformation analysis of Section 5 yields the following nine core regimes. Three carrier-basis pairs contribute one regime each. Three carrier kinds have witnessed split pairs.

The table uses the following short regime labels:

OBL	obligation-bearer identity.
OCC	occurrence identity.
REC	record identity.
LOC	locus-fixed plain-referent identity.
OBJ	object-fixed plain-referent identity.
SCOPE-E	extension-fixed scope identity.
SCOPE-S	structure-fixed scope identity.
RULE-C	content-fixed rule identity.
RULE-S	structure-fixed rule identity.

Their classifications over $\mathcal{F}$ are collected in Table 1. The table records the core profiles used in the lower-bound proof. Three same-carrier cells carry the split arguments: BF for LOC/OBJ, AD for SCOPE-E/SCOPE-S, and RF for RULE-C/RULE-S. Together with carrier disjointness, those cells carry the lower-bound result. The remaining entries summarize diagnostic behavior of the core profiles.

The table should be read as a compact statement of identity behavior. A PRS entry means that the transformation *preserves* the regime’s identity basis. A BRK entry means that the transformation *breaks* that basis and produces a distinct referent under the regime. An NEU entry means that

	OBL	OCC	REC	LOC	OBJ	SCOPE-E	SCOPE-S	RULE-C	RULE-S
RE	NEU	NEU	NEU	NEU	NEU	NEU	NEU	NEU	NEU
AN	NEU	NEU	PRS	NEU	NEU	NEU	NEU	NEU	NEU
RF	PRS	PRS	PRS	PRS	PRS	PRS	BRK	PRS	BRK
AD	PRS	BRK	PRS	PRS	PRS	PRS	BRK	PRS	BRK
RC	NEU	NEU	NEU	NEU	NEU	PRS	BRK	NEU	NEU
RA	NEU	N/A	NEU	N/A	N/A	NEU	BRK	NEU	NEU
SU	BRK	BRK	BRK	BRK	BRK	BRK	BRK	BRK	BRK
BF	NEU	N/A	PRS	PRS	BRK	PRS	BRK	BRK	BRK
PV	NEU	NEU	PRS	NEU	NEU	NEU	NEU	NEU	NEU
SE	NEU	PRS	NEU	PRS	PRS	N/A	N/A	N/A	N/A

Table 1: Classification of each regime profile over the transformation basis $\mathcal{F}$. PRS: preserves identity; BRK: breaks identity; NEU: identity-neutral/non-breaking; N/A: not applicable. Each witnessed split pair has a canonical separating operation: plain referent at BF, scope at AD, and rule at RF. The remaining classifications are diagnostic rather than load-bearing for the lower-bound proof.

the transformation is *identity-neutral* for that regime. An N/A entry means that the transformation is *not applicable* to that regime.

Two columns bracket the profiles. Re-expression is identity-neutral for every core regime: a change of representation format preserves the declared identity basis. Substitution is identity-breaking for every core regime: replacement of the identity carrier breaks identity under each basis. These uniform columns serve as classification sanity checks. They set uniform boundary cases for the profile table. The lower-bound witnesses are the separating cells for the three split pairs.

The witnessed split pairs carry the lower-bound argument. LOC and OBJ are separated by branching or forking. SCOPE-E and SCOPE-S are separated by aggregation or decomposition. RULE-C and RULE-S are separated by refinement. The canonical separating operation explains why each split is forced. The two profiles in a split pair may also differ at other transformations. For scope, re-contextualization and reassignment are two such non-canonical differences. They preserve covered cases and are non-breaking for SCOPE-E. They change applicability context or assignment structure and are breaking for SCOPE-S.

These regimes are defined by identity basis and transformation behavior. They refine identity bases within the existing carrier kinds.

Definition 6.1 (Separation witness). A *separation witness* for identity regimes τ and τ' is a pair of representations (r, s) such that

$$r \sim_{\tau} s \quad \text{and} \quad r \not\sim_{\tau'} s,$$

or conversely.

Proposition 6.2 (Distinct core profiles). *The nine profiles of Table 1 are pairwise distinct.*

Proof. Same-carrier pairs are separated by the split propositions of Section 5. By Proposition 5.7, LOC and OBJ are separated at BF. By Proposition 5.5, SCOPE-E and SCOPE-S are separated at AD. By Proposition 5.6, RULE-C and RULE-S are separated at RF. In each case there is a separation witness: a pair identified under one regime and separated under the other.

Profiles with different carriers govern disjoint sets of representations. A representation of an obligation-bearer is never a representation of an occurrence, an occurrence never a record, a rule never a scope, a scope never a plain referent, and a record never the thing it records. Two regimes whose carriers have disjoint representation domains induce identity relations that differ on those domains, and are therefore distinct by Definition 5.2. Identifying such profiles would collapse the reference inventory of Section 4 and force one carrier to stand in for another.

Therefore every pair of core regimes in Table 1 is separated either by a same-carrier transformation witness or by carrier difference. Appendix A records the separation basis for all $\binom{9}{2} = 36$ pairs. $\square$

7 The Lower Bound

The previous section collected the nine core identity regimes. This section proves the lower-bound claim. The proof requires two facts: the inventory requires stable reference to six carrier kinds, and three of those carrier kinds exhibit witnessed basis plurality under the transformation basis.

7.1 The Inventory Floor

The reference inventory of Section 4 requires stable reference to six kinds of carrier: obligation-bearers, rules, occurrences, scopes, records, and plain referents. Stable reference to a carrier kind requires an identity regime for that kind, because the substrate must determine when representations of that carrier count as the same referent.

Lemma 7.1 (Inventory floor). *Under the assumptions of Section 3, a neutral accountability substrate realizing the inventory of Section 4 must realize at least one identity regime for each required carrier kind.*

Proof. By Definition 4.1, each reference kind in the inventory must be identifiable, trackable, and distinguishable from other referents without making object-level causal or normative commitments. By Definition 3.4, an identity regime specifies sameness for a referent under transformation. Thus each required carrier kind needs identity conditions sufficient to preserve stable reference to carriers of that kind.

The required carrier kinds have disjoint representation domains in the inventory. A representation of an obligation-bearer is not a representation of an occurrence; an occurrence is not a record; a rule is not a scope; a scope is not a plain referent; and a record is not the thing it records. Stable reference to each required carrier kind therefore requires identity conditions for that carrier kind. Collapsing carrier kinds would collapse the reference inventory. Therefore each required carrier kind contributes at least one

identity regime. □

The inventory therefore gives a floor of six regimes. The witnessed basis pluralities for plain referents, scopes, and rules raise that floor to nine.

7.2 Witnessed Splits

Three carrier kinds exhibit witnessed basis plurality under the transformation basis.

First, plain referents split into locus-fixed and object-fixed identity. By Proposition 5.7, branch/fork separates these bases. A fork may preserve the locus while breaking object identity. When both bases are in play, plain referents require both LOC and OBJ.

Second, scopes split into extension-fixed and structure-fixed identity. By Proposition 5.5, aggregation/decomposition separates these bases. A decomposition may preserve the covered cases while changing the internal scope structure. When both bases are in play, scopes require both SCOPE-E and SCOPE-S.

Third, rules split into content-fixed and structure-fixed identity. By Proposition 5.6, refinement separates these bases. A refinement may preserve rule content while changing textual or internal structure. When both bases are in play, rules require both RULE-C and RULE-S.

Each witnessed split adds one regime beyond the one-regime-per-carrier floor. The additional regimes are not new reference kinds. They are distinct identity bases for required carrier kinds.

7.3 The Bound

Theorem 7.2 (Nine-regime lower bound). *Under the assumptions of Section 3, a neutral accountability substrate that supports the inventory of*

Section 4 and supports the witnessed basis pluralities for plain referents, scopes, and rules must realize at least nine identity regimes.

Proof. By Lemma 7.1, the six required carrier kinds in Section 4 require at least six identity regimes.

Three of those carrier kinds require an additional regime when the witnessed basis pluralities are in play. Plain referents require both LOC and OBJ by Proposition 5.7. Scopes require both SCOPE-E and SCOPE-S by Proposition 5.5. Rules require both RULE-C and RULE-S by Proposition 5.6.

Therefore the required core contains:

$$\begin{array}{ccccccc} \text{OBL,} & \text{OCC,} & \text{REC,} & \text{LOC,} & \text{OBJ,} & & \\ \text{SCOPE-E,} & \text{SCOPE-S,} & \text{RULE-C,} & \text{RULE-S.} & & & \end{array}$$

By Proposition 6.2, these nine core profiles are pairwise distinct. Thus the required regimes number at least,

$$1 + 1 + 1 + 2 + 2 + 2 = 9.$$

Thus any neutral accountability substrate satisfying the stated assumptions and supporting the witnessed basis pluralities must realize at least nine identity regimes. This proves the lower-bound claim. $\square$

7.4 Monotonicity

Corollary 7.3 (Monotonicity). *Enlarging the transformation basis cannot lower the bound established by Theorem 7.2.*

Proof. The core is witnessed by distinctions already forced by the transformation basis $\mathcal{F}$. Adding transformations adds classification coordinates. It may leave the existing distinctions unchanged. It may separate an already-distinct pair in further ways. It may also force further splits. But it cannot erase a

split already witnessed by $\mathcal{F}$. Therefore any transformation basis containing $\mathcal{F}$ still requires at least the nine core regimes derived here. $\square$

The monotonicity claim is intentionally limited. It establishes that incompleteness of the transformation basis cannot make the present lower bound disappear. A richer basis may leave the bound unchanged or raise it, but cannot lower it.

7.5 Boundary of the Proof

The claim proved here is a lower bound. Under the stated assumptions and witnessed basis pluralities, fewer than nine regimes cannot preserve stable reference without changing transformation behavior or reintroducing a hidden regime.

The proof does not establish an upper bound. It does not claim that the nine regimes exhaust all coherent identity regimes. It does not claim that every deployed system must expose these regimes. Those broader limits are stated in Section 11.

8 Determinacy

The lower-bound result requires a fixed assignment from core carrier-basis pairs to identity regimes. This section records that assignment and is restricted to the carrier-basis pairs used in the lower-bound construction.

Once a core carrier-basis pair is fixed, the regime assignment is determined. Choosing the identity basis for a real-world referent remains a matter of modeling judgment.

8.1 Core Carrier-Basis Pairs

Definition 8.1 (Core carrier-basis pair). A *core carrier-basis pair* is one of the carrier kind and identity basis combinations used in the lower-bound construction. The core pairs are:

obligation-bearer with responsibility-bearing identity	$\mapsto$	OBL,
occurrence with realization-and-provenance identity	$\mapsto$	OCC,
record with descriptive-record identity	$\mapsto$	REC,
plain referent with locus-fixed identity	$\mapsto$	LOC,
plain referent with object-fixed identity	$\mapsto$	OBJ,
scope with extension-fixed identity	$\mapsto$	SCOPE-E,
scope with structure-fixed identity	$\mapsto$	SCOPE-S,
rule with content-fixed identity	$\mapsto$	RULE-C,
rule with structure-fixed identity	$\mapsto$	RULE-S.

The first three pairs contribute one regime each. The remaining six pairs arise from the witnessed split cases proved in Section 5.

Definition 8.2 (Core regime assignment). The *core regime assignment* maps each core carrier-basis pair to the identity regime displayed in Definition 8.1.

8.2 Existence

Lemma 8.3 (Existence for the core). *Every core carrier-basis pair is assigned to at least one of the nine core regimes.*

Proof. The three one-basis core pairs are assigned by construction: obligation-bearer with responsibility-bearing identity maps to OBL, occurrence with realization-and-provenance identity maps to OCC, and record with descriptive-record identity maps to REC.

The remaining core pairs are assigned by the split results of Section 5. A plain referent with locus-fixed identity maps to LOC, and a plain referent

with object-fixed identity maps to OBJ, by Proposition 5.7. A scope with extension-fixed identity maps to SCOPE-E, and a scope with structure-fixed identity maps to SCOPE-S, by Proposition 5.5. A rule with content-fixed identity maps to RULE-C, and a rule with structure-fixed identity maps to RULE-S, by Proposition 5.6. Therefore every core carrier-basis pair is assigned to at least one of the nine core regimes. $\square$

8.3 Uniqueness

Lemma 8.4 (Uniqueness for the core). *No core carrier-basis pair is assigned to more than one of the nine core regimes.*

Proof. Each core carrier-basis pair contains a fixed carrier kind and a fixed identity basis. The core regime assignment maps that pair to the regime that preserves the declared basis under the transformation behavior used in the lower-bound construction.

The three one-basis core pairs map to OBL, OCC, and REC respectively. For the split cases, the declared basis selects the side of the split: locus-fixed plain referent maps to LOC and not OBJ; object-fixed plain referent maps to OBJ and not LOC; extension-fixed scope maps to SCOPE-E and not SCOPE-S; structure-fixed scope maps to SCOPE-S and not SCOPE-E; content-fixed rule maps to RULE-C and not RULE-S; and structure-fixed rule maps to RULE-S and not RULE-C. Thus the assignment is functional on the core pairs. Therefore no core carrier-basis pair is assigned to more than one core regime. $\square$

8.4 Determinacy Theorem

Theorem 8.5 (Core regime assignment). *Every core carrier-basis pair is assigned to a unique core identity regime.*

Proof. Existence follows from Lemma 8.3. Uniqueness follows from

Lemma 8.4. Therefore every core carrier-basis pair is assigned to a unique core regime. $\square$

This determines the core assignment. It does not decide whether a modeler has chosen the correct identity basis for a particular real-world referent. It says only that the carrier-basis pairs used in the lower-bound construction have fixed regime assignments.

8.5 Failure of Basis Declaration

For the witnessed split cases, omission of the basis is not harmless. A scope without an extension-fixed or structure-fixed basis does not specify whether decomposition preserves covered cases or internal structure. A rule without a content-fixed or structure-fixed basis does not specify whether refinement preserves rule content or textual structure. A plain referent without a locus-fixed or object-fixed basis does not specify whether forking preserves locus or object identity. A declared identity basis is required for deterministic assignment.

This observation anticipates the conformance analysis of the companion *Accountable Records* paper. In a deployed record system, basis declaration becomes a schema obligation. In this paper, it marks the boundary of the determinacy result: regime assignment is fixed for the carrier-basis pairs included in the core.

9 The Nine Regimes at Work

The preceding sections derive the nine core regimes. This section shows how they appear in the running diagnostic settings. The examples are illustrative and do not extend the proof. They show how carrier-basis distinctions arise in ordinary accountability settings.

9.1 Multi-Jurisdiction Regulatory Reporting

A corporation is an obligation-bearer. It persists across jurisdictional differences in corporate personhood, subsidiary responsibility, delegated liability, and compliance theory. It is therefore represented under OBL.

The reporting statute is a rule. If it is identified by its requirements, then it is represented under RULE-C. A later re-codification that preserves the requirements is the same rule under RULE-C because refinement preserves content-fixed identity. If the statute is identified by its section structure, then it is represented under RULE-S. The same re-codification is then a different rule, because refinement changes the structure-fixed identity basis.

Each filing, inspection, transaction, or enforcement action is an occurrence. It is individuated by temporal realization and provenance. It is therefore represented under OCC. Two jurisdictions may disagree about whether the occurrence constitutes compliance. They must still be able to refer to the same occurrence.

The jurisdiction is a scope. If it is identified by the set of regulated cases it covers, then it is represented under SCOPE-E. Splitting the jurisdiction into districts that together cover the same cases preserves identity under SCOPE-E. If it is identified by its named administrative partition, then it is represented under SCOPE-S. The same split changes the structure and therefore breaks identity under SCOPE-S.

Compliance indicators, filings as documents, measurement reports, and audit records are descriptive records. They are represented under REC. They can be compared, annotated, and transferred without the substrate asserting whether compliance actually occurred.

A regulated substance, facility, instrument, parcel, or asset is a plain referent. If the relevant identity basis is the regulated locus, then it is represented under LOC. If the relevant identity basis is the object occupying or moving through that locus, then it is represented under OBJ.

The case exercises all nine core regimes. Two jurisdictions can refer to the same corporation, statute, filing, jurisdiction, record, and facility while evaluating compliance incompatibly. They can also disagree about whether decomposing a jurisdiction changed the jurisdiction. That disagreement is the SCOPE-E/SCOPE-S distinction. A count that stops at the reference kind “scope” would miss it.

9.2 Legislative History Tracking

A statute is a rule. If tracked by normative content, it is represented under RULE-C. An amendment or refinement that preserves the statutory requirements preserves identity under RULE-C. If tracked by text, section structure, or internal organization, the statute is represented under RULE-S. The same amendment or refinement may then produce a new structure-fixed rule.

The enactment, amendment, vote, administrative application, judicial decision, or repeal event is an occurrence. It is represented under OCC. The occurrence may enact, modify, interpret, or challenge the statute, but it is not identical with the statute.

The legislature, agency, court, officer, or regulated party is an obligation-bearer when represented as a party that can bear responsibility, authority, or accountability. Such entities are represented under OBL. Their identity does not depend on any single enactment or application event.

Geographic, subject-matter, procedural, temporal, or institutional boundaries are scopes. A statutory scope may be tracked extensionally by the cases it covers, yielding SCOPE-E. It may also be tracked structurally by the named jurisdictional or procedural partition, yielding SCOPE-S. Redistricting, reclassification, or subdivision can preserve coverage while changing structure.

Legislative records, committee reports, interpretive memoranda, docket entries, and compliance histories are records. They are represented under

REC. They may record that someone asserted a causal or normative claim without making that claim at the substrate level.

Objects acted upon by the legal regime, such as land parcels, facilities, permits, instruments, samples, or regulated assets, are plain referents. They may be locus-fixed under LOC or object-fixed under OBJ, depending on the declared identity basis.

The case shows why the rule split matters. Legal practice often needs both readings. A doctrine may remain unchanged while the text is superseded. A text may remain identifiable while its content is interpreted differently. The RULE-C/RULE-S split keeps those identity bases distinct.

9.3 Cross-Institutional Provenance

Cross-institutional provenance separates records from the things those records describe. A dataset entry, measurement record, transfer log, annotation record, or derived metric is represented under REC. It may be copied, annotated, refined, and extended with provenance while remaining the same record.

The sampled object, monitored site, instrument, facility, specimen, or collection point is a plain referent. If the identity basis is the site, location, or collection point, then it is represented under LOC. A forked representation over the same site preserves the locus-fixed referent. If the identity basis is the specimen, instrument, or physical object, then it is represented under OBJ. A fork that produces distinct object continuations breaks object-fixed identity.

Producing institutions, custodians, laboratories, agencies, and receiving organizations are obligation-bearers when represented as parties responsible for custody, transfer, use, retention, or analysis. They are represented under OBL.

Data-use policies, retention rules, consent terms, sharing agreements, and

access restrictions are rules. When tracked by policy content, they are represented under RULE-C. When tracked by their text, version, or internal structure, they are represented under RULE-S.

Transfers, annotations, re-analyses, approvals, denials, measurements, and custody changes are occurrences. They are represented under OCC. They remain referable as the same events even when institutions disagree about their significance.

Institutional coverage conditions, consent scopes, policy domains, project boundaries, and data-sharing jurisdictions are scopes. They may be extension-fixed under SCOPE-E or structure-fixed under SCOPE-S. A policy domain may cover the same records after decomposition while no longer having the same internal structure.

The provenance chain can therefore remain stable while institutions disagree about classification, causal interpretation, data quality, responsibility, or permissible use. The LOC/OBJ distinction keeps “the same site” and “the same specimen” from being conflated across a fork. The record-versus-referent distinction keeps a record from being confused with the thing it records.

9.4 Summary

The proof of the lower bound was given in Section 7. The running cases show that the core distinctions correspond to ordinary record-system pressures.

Each case uses the three core regimes that are not split in this construction: OBL, OCC, and REC. Each case also exercises the three witnessed split pairs: LOC/OBJ, SCOPE-E/SCOPE-S, and RULE-C/RULE-S. The additional regimes correspond to identity distinctions that ordinary record operations already force.

A jurisdiction that stays the same under decomposition and one that does not are both scopes. A statute individuated by its requirements and a statute individuated by its text are both rules. A site and the object occupying it

may both be plain referents. The kind names are the same. The identity bases are not.

Reference kinds supply carriers. Transformations determine regimes. Counting reference kinds gives six. The transformation analysis forces at least nine identity regimes.

10 Related Work

This paper is situated at the intersection of formal ontology, provenance, and classification under disagreement. It does not propose a new upper ontology. It derives a lower bound on identity regimes required when a substrate is optimized for neutral reference under persistent disagreement.

10.1 Classification under Disagreement

Classification systems embed institutional, social, and evaluative commitments rather than functioning as neutral descriptions (Bowker and Star 1999; Haslanger 2012). In scientific and policy settings, disagreement is often persistent rather than transient, reflecting incompatible evidentiary, methodological, or evaluative frameworks rather than temporary empirical uncertainty (Longino 1990). This motivates representations that remain usable without presupposing interpretive convergence.

This paper adopts that motivation but addresses a narrower structural question: which identity distinctions become unavoidable when disagreement is persistent and causal or normative commitment is externalized from the substrate. The answer is conditional on the neutral-substrate constraint developed in *Neutral Substrates* (Case 2026) and on the formal setting fixed above. Under those assumptions, stable reference requires at least the nine core identity regimes derived here.

10.2 Identity, Roles, and Ontological Methodology

Formal ontology has long emphasized that categories are individuated by identity conditions and persistence criteria. OntoClean clarifies why anti-rigid properties such as roles cannot supply identity conditions (Guarino 1999; Guarino and Welty 2002). Work on social and institutional entities further develops distinctions among bearers, roles, social objects, and institutional structures (Masolo et al. 2003; Guizzardi 2005).

This paper builds on those insights but applies them at the substrate level. Its regimes are not derived from domain taxonomy. They are derived from transformation behavior over identity bases. The central move is to make identity operational: a regime is determined by whether ordinary transformations preserve, break, or leave identity neutral.

10.3 Upper Ontologies

Upper ontologies such as BFO, DOLCE, and UFO provide rich accounts of dependence, endurants and perdurants, social objects, qualities, and institutional entities (Smith et al. 2015; Arp et al. 2015; Gangemi et al. 2002; Guizzardi 2005). They support broad ontology-engineering objectives, including reuse, domain modeling, semantic integration, and metaphysical clarity. The present result addresses a narrower design objective: stable substrate-level reference under persistent disagreement. It offers a constraint that can be imposed on implementations when that objective is adopted.

From the perspective of BFO, several of the required carriers have natural representational starting points. Occurrences map naturally to occurrents. Material plain referents map naturally to independent continuants. Rules, records, roles, dispositions, information artifacts, and institutional bearers can also be modeled using BFO resources, depending on the intended application. When a BFO-based implementation is intended to serve as a neutral substrate under persistent disagreement, the relevant identity bases can be

specified as part of the implementation. For example, non-spatial normative applicability, record identity, and obligation-bearing capacity may require dedicated modeling patterns to ensure that the regime-level distinctions are preserved across transformations.

DOLCE and UFO provide rich resources for social objects, institutional entities, qualities, roles, and dependence relations. These resources make them natural candidates for representing obligation-bearers, rules, and institutional structures. Here again, the contribution is an additional constraint imposed when neutrality under persistent disagreement is adopted as a design objective. A deployed substrate must make clear when it is tracking a rule by content rather than structure, a scope by extension rather than internal organization, a plain referent by locus rather than object, and a record as distinct from the thing recorded. Where those distinctions are handled by local modeling conventions, role predicates, or application logic, the implementation may consider whether those conventions remain stable when shared interpretation cannot be assumed.

These observations are illustrative. The core is derived from the transformation analysis of Sections 5–7, not from any gaps in upper ontologies. Upper ontologies can realize the regime structure derived here when their implementations preserve the relevant identity behavior. The benefit is a reusable constraint for cases where an ontology must support neutral reference across divergent interpretations.

10.4 Provenance, Versioning, and Exchange

The transformation basis used in this paper mirrors operations already tracked by provenance and versioning systems. PROV represents derivation, attribution, generation, use, and revision in ways that overlap naturally with re-expression, annotation, refinement, branching, and provenance extension (Moreau and Missier 2013). This paper adds an identity-level constraint for cases where provenance and versioning records must remain referable

across incompatible interpretations.

The neutral substrate and its regimes can be stated as a conservative theory over an exchange formalism such as Common Logic (ISO/IEC 2018). Conservativity matters because extensions should add interpretive structure without changing consequences about substrate-level terms (Hodges 1997). This supports heterogeneous ontology exchange and pluralistic structuring in settings where no single interpretive framework can govern all use (Kutz et al. 2010).

Large-scale information infrastructures and digital-twin systems face related problems of long-lived reference, versioning, and cross-institutional reuse (Edwards 2010; Grieves and Vickers 2017). The literature emphasizes that infrastructure must remain usable across evolving communities and interpretations. This paper provides a constraint that enables stable reference under persistent disagreement.

10.5 Causal and Normative Separation

The distinction between descriptive record and causal claim follows a familiar discipline in causal analysis. Causal claims require additional modeling commitments and should not be confused with descriptive structure alone (Pearl 2009). The neutral-substrate constraint applies the same discipline at the substrate boundary. A substrate may record that a source asserted a causal or normative proposition. It must not thereby assert the proposition as substrate-level fact.

The regimes derived here make that separation usable under transformation. Records can persist, be annotated, forked, and extended with provenance without becoming causal explanations or normative judgments. Rules can remain referable without the substrate deciding which interpretation governs. Occurrences can remain referable without the substrate deciding their causal effects or compliance status. That separation is essential when identity must remain stable before interpretation is settled.

10.6 FAIR Reuse under Persistent Disagreement

The FAIR principles promote data reuse across communities and over time (Wilkinson et al. 2016). They do not specify the ontological identity conditions required when reuse occurs under persistent legal, institutional, or analytic disagreement. In practice, reuse is often pursued through negotiated vocabularies, shared schemas, or domain-specific interpretive conventions. Those strategies are valuable, but they presuppose forms of convergence that may be unavailable in contested settings.

This result complements FAIR-aligned reuse in contested settings. Stable reference must be maintained independently of causal or normative agreement. The nine core regimes name identity distinctions that a neutral substrate must represent when reuse is expected to continue across divergent interpretations.

11 Limits and Boundary Conditions

The result of this paper is deliberately narrow. It applies to neutral accountability substrates satisfying the assumptions of Section 3. It establishes a necessary lower bound on identity regimes: a floor, not a ceiling. It does not certify that a deployed system has chosen the correct regime for every real-world referent.

11.1 Conditional Scope

The lower bound depends on the neutral-substrate constraint developed in *Neutral Substrates* (Case 2026) and on the formal setting fixed in this paper. In particular, it assumes persistent disagreement, strong stable reference, minimality, and exclusion of hidden regimes. If those assumptions are rejected, the bound need not apply.

The result is structural rather than domain-specific. It concerns identity behavior under transformation. It does not depend on any particular legal

theory, policy framework, institutional vocabulary, or causal model. It says that, under the stated assumptions, fewer than nine core regimes cannot preserve stable reference for the carrier-basis pairs analyzed here without either changing identity behavior or hiding a regime distinction elsewhere.

11.2 No Upper Bound or Completeness Claim

The nine-regime core is a lower bound. It leaves three questions open: whether these nine regimes exhaust the coherent identity regimes; whether the transformation basis $\mathcal{F}$ is complete for all possible record systems; and whether some domains require additional reference kinds.

The monotonicity result of Corollary 7.3 is one-directional. Adding transformations cannot collapse distinctions already witnessed by $\mathcal{F}$. A richer basis may leave the bound unchanged or force further splits. It cannot make the nine-regime lower bound disappear.

The inventory supplies the carrier kinds required for the lower-bound construction. It does not by itself determine identity. Identity is fixed when a carrier kind is paired with an identity basis and evaluated under the transformation basis.

The paper does not claim that obligation-bearers, occurrences, or records admit no further admissible identity bases. For example, a record may be tracked by descriptive content or by a particular carrier, file, signature, or chain-of-custody token. An occurrence may be tracked as a fine-grained realization or as a coarser event complex. An obligation-bearer may be tracked as a legal person, office-holder, delegated role, or responsible organizational unit. If such bases are required by a deployment, they may force further regimes. That possibility does not weaken the lower bound. It confirms that the present result is a lower bound, not a completeness theorem.

11.3 Excluded Systems

Several classes of systems fall outside the scope of the result.

Interpretively dependent identity. A system is outside the scope if the identity condition of a referent depends on causal, normative, evaluative, or methodological commitments supplied by an extension. Such a system may be useful within a coordinated framework. It is not a neutral substrate in the sense analyzed here.

Relation-determined identity. A system is outside the scope if relations introduce, modify, or determine the identity conditions of their relata. Relations may connect already-individuated referents. They may not supply the identity basis that makes those referents the same or different.

Extension-variant identity. A system is outside the scope if identity conditions vary when admissible extensions are added. Such systems violate the stable-reference requirement. They may support local or negotiated identity. They do not support the strong form of stable reference studied here.

Embedded interpretive commitments. A system is outside the scope if it makes substrate-level causal or normative commitments whose validity depends on a particular interpretive framework. The substrate may record attributed causal or normative claims. It may not assert them as object-level substrate-layer commitments.

Undeclared hidden regimes. A system is outside the scope if it collapses regimes and then recovers the lost distinction through roles, flags, contextual predicates, workflow states, schema conventions, or application-level discrim-

inators. Such a system has not reduced structure. It has relocated structure into an interpretive mechanism.

Unmodeled transformation behavior. A system is outside the scope when its essential identity behavior depends on transformations not represented in $\mathcal{F}$ and not reducible to the transformation families used here. Such a case does not refute the lower bound. It indicates that the transformation basis should be enlarged. By monotonicity, such enlargement may preserve the bound or force further regimes.

11.4 Non-Goals and Weakened Objectives

Several questions remain outside this paper. This paper constructs no causal or normative extensions. It takes no position on which causal account is correct or which normative interpretation should govern. It offers no domain-specific modeling guidance. It does not claim that all useful ontologies should be neutral substrates. It does not assess the use of hidden regimes except to note that hidden regimes are incompatible with the specific objective analyzed here: stable substrate-level reference under persistent disagreement.

If stability is weakened to permit context-dependent regime membership, some distinctions may be represented as role-relative or parameter-dependent refinements of fewer base types. That may be a reasonable engineering choice in coordinated settings. But then the structural burden moves into coordination, versioning, governance, or application logic whose admissibility is not fixed at the substrate level. The lower bound proved here does not apply under that weakened objective.

11.5 Boundary with Accountable Records

This paper concerns identity. It derives the lower-bound regime structure required for stable reference. It does not provide a conformance checker for

deployed record systems. It does not specify how schemas must declare bases, logs, or discriminator surfaces. It does not decide whether an implementation has disclosed every identity-relevant surface.

Those questions are deferred to the companion *Accountable Records* paper. That paper is intended to develop accountable records, basis declaration, transformation logs, and conformance checks. The present paper supplies the lower-bound identity core that such a conformance analysis requires. It establishes part of the identity cost of the neutral-substrate constraint, while *Accountable Records* studies how that cost can be made checkable in deployed record systems.

11.6 Boundary of the Claim

Within the stated scope, the conclusion is unavoidable. A neutral accountability substrate that supports the inventory of Section 4, preserves stable reference under the transformation basis $\mathcal{F}$, and excludes hidden regimes must realize at least the nine core identity regimes. Outside that scope, different optimization objectives may yield different structures. The result is best understood as a constraint theorem: for neutral substrate-level reference under persistent disagreement, at least this much identity structure is required.

12 Conclusion

Naming kinds does not fix identity in a neutral substrate. It is fixed by specifying the transformations that preserve sameness. Once the ordinary operations of accountability systems are specified, individuation depends on transformation behavior.

This paper began with the neutral-substrate constraint developed in *Neutral Substrates* (Case 2026). A substrate that must remain usable under persistent

disagreement may commit to stable reference and permitted attribution, but not to object-level causal or normative interpretation. This paper derives the identity structure required by that constraint when records are transformed in ordinary use.

The result is a nine-regime lower bound. An accountability substrate must be able to refer to six carrier kinds: obligation-bearers, rules, occurrences, scopes, records, and plain referents. Those six reference kinds supply carriers, but do not settle identity. Three of them are assigned more than one admissible basis: a scope by extension or by structure, a rule by content or by structure, a plain referent by locus or by object. Ordinary transformations force apart those alternatives: decomposition separates extension-fixed from structure-fixed scopes, refinement separates content-fixed from structure-fixed rules, and forking separates locus-fixed from object-fixed plain referents. Thus six reference kinds force at least nine identity regimes:

OBL, OCC, REC, LOC, OBJ,
SCOPE-E, SCOPE-S, RULE-C, RULE-S.

The central result is the identification of the additional regimes. They are not additional reference kinds, but distinct identity bases forced apart by transformation behavior. An analysis that names kinds may undercount the structure required for stable reference. Collapsing regimes does not produce genuine neutrality. A collapsed distinction must change identity behavior or be recovered through a hidden regime.

The result is conditional. It holds under the assumptions fixed in this paper: persistent disagreement, strong stable reference, minimality, and exclusion of hidden regimes. It is a lower bound rather than an upper bound or a completeness claim. A richer transformation basis may leave it unchanged or force further regimes.

The result is diagnostic rather than prescriptive. It does not claim that every ontology should expose these nine regimes, nor that contextual typing, role

predicates, or hidden discriminators are useless as engineering devices. It says that such devices cannot substitute for regime-level identity commitments in a neutral substrate under persistent disagreement. When stable reference must be held as common ground before interpretation is settled, the substrate structure must preserve the relevant identity distinctions.

Three papers in this series explore the implications of that constraint. *Neutral Substrates* states the neutrality constraint. *Referential Regimes* measures its identity cost. *Accountable Records* studies how that cost can be made checkable in deployed record systems. This work addresses the middle step: stable reference under persistent disagreement requires transformation-invariant identity regimes, and at least nine such regimes are forced by the ordinary operations accountability systems already perform.

Statements and Declarations

Acknowledgments

The author thanks the anonymous reviewers of an earlier version of this work for comments that helped improve the framing, organization, and presentation of the work.

Author Contributions

The author is the sole contributor to this work and is responsible for all aspects of the research, authorship, and publication.

Use of AI-Assisted Tools

AI-assisted tools were used for editing, formatting, and consistency checking. The author reviewed all suggestions and is solely responsible for the content.

Declaration of Conflicting Interest

The author declares no potential conflicts of interest with respect to the research, authorship, and publication of this work.

A Pairwise Separation

The nine regimes yield $\binom{9}{2} = 36$ unordered pairs. Each pair is separated either by carrier difference or, for same-carrier split pairs, by a separation witness in the sense of Definition 6.1. Three pairs share a reference kind and differ by identity basis: LOC/OBJ, SCOPE-E/SCOPE-S, and RULE-C/RULE-S. All other pairs are separated by carrier difference.

	OBL	OCC	REC	LOC	OBJ	SCOPE-E	SCOPE-S	RULE-C	RULE-S
OBL	–	•	•	•	•	•	•	•	•
OCC	•	–	•	•	•	•	•	•	•
REC	•	•	–	•	•	•	•	•	•
LOC	•	•	•	–	BF	•	•	•	•
OBJ	•	•	•	BF	–	•	•	•	•
SCOPE-E	•	•	•	•	•	–	AD	•	•
SCOPE-S	•	•	•	•	•	AD	–	•	•
RULE-C	•	•	•	•	•	•	•	–	RF
RULE-S	•	•	•	•	•	•	•	RF	–

Here • indicates separation by carrier difference. The entries BF, AD, and RF indicate same-kind split pairs separated by the corresponding canonical operation.

The LOC/OBJ pair is separated by branch/fork. A fork over a fixed locus may preserve locus identity while breaking object identity. Thus the two regimes induce distinct identity relations.

The SCOPE-E/SCOPE-S pair is separated by AD. A whole-preserving decomposition may preserve the covered cases while changing the internal scope structure. Thus the two regimes induce distinct identity relations.

The RULE-C/RULE-S pair is separated by refinement. A refinement may preserve rule content while changing textual or internal rule structure. Thus the two regimes induce distinct identity relations.

Therefore every pair of regimes is separated either by carrier difference or by a transformation witness. No pair collapses.

References

- R. Arp, B. Smith, and A. D. Spear. *Building Ontologies with Basic Formal Ontology*. MIT Press, 2015.
- G. C. Bowker and S. L. Star. *Sorting Things Out: Classification and Its Consequences*. MIT Press, Cambridge, MA, 1999.
- D. M. Case. *Neutral Substrates: A Design Constraint for Shared Records Under Persistent Interpretive Disagreement*. arXiv preprint arXiv:2601.14271, 2026.
- P. N. Edwards. *A Vast Machine: Computer Models, Climate Data, and the Politics of Global Warming*. MIT Press, 2010.
- A. Gangemi, N. Guarino, C. Masolo, A. Oltramari, and L. Schneider. Sweetening ontologies with DOLCE. In *Knowledge Engineering and Knowledge Management: Ontologies and the Semantic Web (EKAW 2002)*, volume 2473 of LNCS, pages 166–181. Springer, 2002.
- M. Grieves and J. Vickers. Digital twin: Mitigating unpredictable, undesirable emergent behavior in complex systems. In *Transdisciplinary Perspectives on Complex Systems*, pages 85–113. Springer, 2017.
- N. Guarino. The role of identity conditions in ontology design. In C. Freksa and D. M. Mark, editors, *Spatial Information Theory: Cognitive and Computational Foundations of Geographic Information Science (COSIT '99)*, volume 1661 of LNCS, pages 221–234. Springer, 1999.
- N. Guarino and C. A. Welty. Evaluating ontological decisions with OntoClean. *Communications of the ACM*, 45(2):61–65, 2002.
- G. Guizzardi. *Ontological Foundations for Structural Conceptual Models*. PhD thesis, University of Twente, 2005.
- S. Haslanger. *Resisting Reality: Social Construction and Social Critique*. Oxford University Press, 2012.

- W. Hodges. *A Shorter Model Theory*. Cambridge University Press, 1997.
- ISO/IEC. Information technology — Common Logic (CL): A framework for a family of logic-based languages. ISO/IEC 24707:2018, 2018.
- O. Kutz, T. Mossakowski, and D. Lücke. Carnap, Goguen, and the hyperontologies: Logical pluralism and heterogeneous structuring in ontology design. *Logica Universalis*, 4(2):255–333, 2010.
- H. E. Longino. *Science as Social Knowledge: Values and Objectivity in Scientific Inquiry*. Princeton University Press, 1990.
- C. Masolo, S. Borgo, A. Gangemi, N. Guarino, and A. Oltramari. WonderWeb Deliverable D18: Ontology Library. IST Project 2001-33052 WonderWeb, 2003.
- L. Moreau and P. Missier, editors. *PROV-DM: The PROV Data Model*. W3C Recommendation, 30 April 2013.
- J. Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2nd edition, 2009.
- B. Smith et al. *Basic Formal Ontology 2.0: Specification and User’s Guide*. Technical specification, 26 June 2015.
- M. D. Wilkinson et al. The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, 3:160018, 2016.